

Master's Thesis

Agentic AI for CrypTool-Online Developers

Konzepte zur produktiven Nutzung von
LLMs (AI Agent) in der Entwicklung von
großen Softwareprojekten am Beispiel von
CrypTool

Concepts for the Productive use of LLMs
(AI Agents) in the Development of Large
Software Projects Using the Example of
CrypTool

Examiner Prof. Bernhard Esslinger
Supervisor Felix Breitweiser

Author Michael Heep
Submitted tbd

Contents

1	Introduction	5
1.1	Motivation and Problem Statement	5
1.2	Research Questions	5
1.3	Delimitations	5
1.4	Structure	5
2	Background and Related work	5
3	Theoretical Foundations	5
3.1	Agentic AI	5
4	Methodology	5
5	Implementation and Case Studies	5
5.1	Claude Code	5
5.2	Free Models	5
6	Results and Discussion	5
7	Conclusion and Future Work	5

1 Introduction

1.1 Motivation and Problem Statement

1.2 Research Questions

1.3 Delimitations

1.4 Structure

2 Background and Related work

3 Theoretical Foundations

3.1 Agentic AI

4 Methodology

5 Implementation and Case Studies

5.1 Claude Code

5.2 Free Models

6 Results and Discussion

7 Conclusion and Future Work